THE COMPLETE CASE STUDY MASTERY GUIDE

Data Science & Analytics Interviews — From Zero to Superpower

---

PART I: FOUNDATIONS — HOW CASE STUDIES ACTUALLY WORK

Chapter 1: What Interviewers Are Really Evaluating

Most candidates believe case studies test whether you can get the "right answer." This is fundamentally wrong. Case studies test **how you think**, not what you conclude. Two candidates can reach opposite recommendations and both get hired — or both get rejected — based entirely on the quality of their reasoning.

The 5 Evaluation Axes

Every interviewer, whether they realize it or not, is scoring you on five dimensions:

Axis 1: Structuring Ability (Weight: ~35%)

This is the single most important skill. Can you take a vague, messy, real-world problem and decompose it into a clean, logical framework? The interviewer is watching whether you jump straight into analysis (bad) or pause to organize your thinking first (good).

What "good" looks like: "Before I dive in, let me break this problem into three areas. First, I want to understand whether this is a supply-side or demand-side issue. Second, I want to identify which user segments are most affected. Third, I want to examine whether this is a product change or an external factor."

What "bad" looks like: "So I think what's happening is that users probably don't like the new feature, so I'd run an A/B test to see if that's the case, and also maybe look at the data to see if there's a trend..."

Axis 2: Business Intuition (Weight: ~25%)

Do you understand how the business works? Do you know what metrics actually matter for a subscription company vs. a marketplace vs. an ad-supported platform? Can you connect a data observation to a business consequence?

This is where many technically strong candidates fail. They can run a regression but can't explain why a 5% drop in Day-7 retention matters more than a 10% drop in page views.

To build this: Read 10-K filings of major tech companies. Understand their revenue models. Know the difference between a company that monetizes through ads (Meta, Google), subscriptions (Netflix, Spotify), transactions (Uber, DoorDash), or enterprise contracts (Slack, Salesforce). Each model produces completely different "important" metrics.

Axis 3: Analytical Reasoning (Weight: ~20%)

This is NOT "can you code in SQL" or "can you build a model." It is: given a problem, can you identify the RIGHT analysis to run? Can you reason about what the data would look like if Hypothesis A is true versus Hypothesis B? Can you design an analysis that actually answers the question?

Example of strong analytical reasoning: "If this revenue drop is caused by a mix shift toward lower-value users, I'd expect to see revenue-per-user holding steady within each segment but the proportion of high-value users declining. If instead it's caused by across-the-board price sensitivity, I'd expect revenue-per-user declining in every segment."

Axis 4: Communication (Weight: ~15%)

Can you explain your thinking clearly and concisely? Do you use signposting ("There are three possible causes — let me walk through each")? Can you summarize without rambling? Do you answer the question that was asked?

The bar here is: could your interviewer turn around and re-explain your analysis to their VP? If your explanation is too convoluted for that, you've failed this axis.

Axis 5: Technical Depth (Weight: ~5%, but it's a tiebreaker)

When the interviewer probes ("How would you actually test that hypothesis?"), can you go deep? Can you discuss sample size calculations, the right statistical test, potential confounders, or why a particular ML model would or wouldn't work here?

This axis rarely determines the outcome alone, but it separates "strong hire" from "hire" in close calls.

Chapter 2: The 7 Case Archetypes

Every case study you will ever encounter in a data science or analytics interview maps to one of seven archetypes. Some cases combine two archetypes, but the core patterns are always drawn from this list. Memorize these. When you hear a case prompt, your first mental move should be: "Which archetype is this?"

Archetype 1: Metric Diagnosis

The prompt pattern: "X metric went up/down by Y%. Why? What would you do?"

Examples:

- "Daily active users on our app dropped 15% last week."
- "Revenue per user increased 8% but total revenue is flat."
- "Our app's crash rate spiked 3x on Tuesday."

Why it's the most common: ~40-50% of all case interviews are metric diagnosis. Companies deal with metric movements constantly, and they want to know if you can systematically diagnose a problem rather than guessing.

The core skill tested: Can you decompose a metric into its components, segment the data to isolate the problem, and generate prioritized hypotheses?

Archetype 2: Metric Definition

The prompt pattern: "How would you measure the success of X?"

Examples:

- "We just launched a Stories feature. How do you measure if it's successful?"
- "What should the North Star metric be for our food delivery app?"
- "How would you build a health score for enterprise customer accounts?"

The core skill tested: Can you connect business goals to measurable metrics? Do you understand the difference between vanity metrics and actionable metrics? Can you define guardrail metrics?

Archetype 3: Product Sense / Feature Evaluation

The prompt pattern: "Should we build X?" or "How would you improve X?"

Examples:

- "Should Instagram add a dislike button?"

- "How would you improve the Uber rider experience?"
- "We're considering launching a premium tier. Walk me through your analysis."

The core skill tested: Can you think about products from the user's perspective AND the business perspective simultaneously? Can you identify tradeoffs?

Archetype 4: A/B Test Design and Analysis

The prompt pattern: "Design an experiment to test X" or "Here are some A/B test results — interpret them."

Examples:

- "We want to test a new checkout flow. Design the experiment."
- "We ran an A/B test and the results show a 3% lift in clicks but a 1% drop in purchases. What do you do?"
- "Our A/B test has been running for 2 days and shows p=0.03. Should we ship?"

The core skill tested: Do you understand experimental design, statistical significance, practical significance, and common pitfalls like peeking, novelty effects, and network interference?

Archetype 5: Root Cause Analysis (Multi-Variable)

The prompt pattern: "Metric A went up but Metric B went down. What's happening?"

Examples:

- "Signups are up 20% but activation rate dropped 15%."
- "Customer satisfaction scores improved but churn increased."
- "Page views are up but ad revenue per impression is down."

The core skill tested: Can you reason about relationships between metrics? Can you identify confounding variables, mix shifts, and Simpson's Paradox?

Archetype 6: Growth and Strategy

The prompt pattern: "How would you grow X?" or "What strategy would you recommend for Y?"

Examples:

- "How would you grow Spotify's user base in Southeast Asia?"
- "What's your strategy for reducing churn by 25% this quarter?"
- "We have $5M in marketing budget. How would you allocate it?"

The core skill tested: Can you think systematically about growth levers? Do you understand acquisition funnels, retention curves, and unit economics? Can you prioritize initiatives by expected impact?

Archetype 7: Data Modeling and Pipeline Design

The prompt pattern: "Design a data schema for X" or "How would you build Y dashboard/model?"

Examples:

- "Design the data tables needed for a recommendation engine."
- "Build a dashboard to monitor marketplace health."
- "How would you set up a churn prediction pipeline?"

The core skill tested: Can you translate business requirements into data architecture? Do you understand star schemas, event logging, ETL processes, and how analytics teams actually work with data?

Chapter 3: The SPADE Framework — Your Universal Skeleton

Every case, regardless of archetype, follows the same high-level structure. The SPADE framework gives you a reliable starting point so you never freeze or ramble. With practice, this structure becomes automatic.

S — Situation (First 30-60 seconds)

What to do: Restate the problem in your own words to confirm you understood it correctly. Then ask 2-3 smart clarifying questions.

Why this matters: Restating shows you're listening. Clarifying questions show you're thinking critically before you start analyzing. They also buy you precious thinking time.

What makes a clarifying question "smart":

- It changes your approach depending on the answer
- It shows you understand the business context
- It eliminates ambiguity that could lead you down the wrong path

Good clarifying questions:

- "When you say DAU dropped, is that the total count or the ratio to MAU? And over what time period — sudden drop or gradual decline?" (This changes whether you investigate a one-time event vs. a systemic trend.)
- "Is this across all platforms or specific to mobile/web?" (This narrows your search space immediately.)
- "Was there any recent product release, pricing change, or marketing campaign that coincides with this change?" (This checks the most common cause first.)

Bad clarifying questions:

- "What does DAU mean?" (You should know this.)
- "Can you tell me more about the company?" (Too vague. Shows no initiative.)
- "What do you think the cause is?" (Never ask the interviewer to do your job.)

P — Problem Decomposition (Minutes 1-3)

What to do: Break the problem into 3-5 mutually exclusive, collectively exhaustive (MECE) sub-components. Present them as a structured tree.

MECE explained: "Mutually exclusive" means the categories don't overlap — an issue can only fall into one bucket. "Collectively exhaustive" means the categories cover all possibilities — nothing is missed. This is the backbone of structured thinking.

Example for "Revenue dropped 10%":

Revenue = Number of Users × Average Revenue Per User (ARPU)

Branch 1: Did the number of users decline?

- New user acquisition down?
- Existing user churn up?
- Reactivation of dormant users down?

Branch 2: Did ARPU decline?

- Are users purchasing less frequently?
- Is the average transaction value lower?
- Did pricing or discounting change?

Branch 3: External factors

- Competitive landscape change?
- Seasonal pattern?
- Macroeconomic conditions?

Branch 4: Data/technical issues

- Logging or tracking changes?
- Data pipeline delays?
- Metric definition change?

The power of decomposition: Notice how this structure prevents you from guessing. Instead of saying "Maybe users don't like the product anymore," you systematically check each branch. This is what separates a structured thinker from a hand-waver.

A — Analysis Plan (Minutes 3-6)

What to do: For each branch of your decomposition, describe what data you would pull, what analysis you would run, and what result would confirm or reject each hypothesis.

Template for each branch:

"For [Branch X], I would pull [specific data] and look for [specific pattern]. If I see [Result A], that would suggest [Hypothesis A is true]. If instead I see [Result B], I can rule this out and move to the next branch."

Example continuing the revenue case:

"For Branch 1 — user count decline — I'd start by looking at a cohort analysis. I'd break users into new, retained, and resurrected segments and plot each over the last 8 weeks. If I see new user acquisition dropped significantly while retention held steady, that points to a top-of-funnel problem — maybe a marketing channel underperformed or a competitor launched. If instead retention dropped while new user acquisition held, that's a product or experience problem."

"For Branch 2 — ARPU decline — I'd look at the distribution of revenue per user over time. Is the median dropping, or is it the mean being pulled down by a change in the tail? I'd also check whether there was a pricing change, a shift in product mix, or an increase in discount usage."

D — Data and Decision (Minutes 6-9)

What to do: Walk through what you expect the data to show based on the most likely hypothesis. If the interviewer gives you actual data, analyze it. Then make a clear recommendation.

How to handle "what if" data: Sometimes interviewers will say "Let's say you pulled the data and found X." Roll with it. Interpret X, explain what it implies, and adjust your framework accordingly.

Making a recommendation:

- State your conclusion clearly: "Based on this analysis, the most likely root cause is..."
- Quantify the impact if possible: "This segment accounts for approximately 40% of total revenue, which would explain about 4 of the 10 percentage points of decline."

- Recommend specific actions: "I'd recommend three actions in priority order..."
- State your confidence level: "I'm fairly confident in this diagnosis, though I'd want to validate by checking..."

E — Edge Cases and Extensions (Minutes 9-10)

What to do: Proactively address limitations, risks, and what you'd do next. This is where you differentiate yourself from other candidates.

Things to mention:

- "One risk of this recommendation is... I'd mitigate that by..."
- "A limitation of this analysis is that it doesn't account for... To address this, I'd also look at..."
- "After implementing the fix, I'd monitor [specific metrics] daily for the next two weeks to confirm recovery."
- "One second-order effect to watch for is..."
- "If I had more time, I'd also investigate... because..."

PART II: METRIC DIAGNOSIS — THE MOST IMPORTANT SKILL

Chapter 4: The Metric Decomposition Engine

Metric diagnosis is the single most important case type to master. It appears in nearly half of all interviews, and the skills it builds (decomposition, segmentation, hypothesis generation) transfer directly to every other case type.

The Fundamental Principle: Every Metric Is a Formula

The first thing you must do when you hear a metric name is mentally convert it into a mathematical formula. This is non-negotiable. If you can't decompose the metric, you can't diagnose it.

Master Decomposition List — Memorize These:

Revenue:

Revenue = Users × Transactions per User × Revenue per Transaction

Alternatively:

Revenue = Volume × Price
Revenue = Paying Users × ARPU
Revenue = Impressions × CTR × CPC (for ad revenue)

Daily Active Users (DAU):

DAU = New Users + Retained Users + Resurrected Users − Churned Users

This is critical. A DAU drop could come from any of these four components. Most candidates only think about new users and churn — they forget resurrected users entirely.

Conversion Rate (for any funnel):

Overall Conversion = Step1→Step2 × Step2→Step3 × Step3→Step4 × ...

Always break the funnel into individual steps. A conversion drop almost always concentrates in one specific step, not across the whole funnel.

Engagement:

Engagement = Sessions per User × Actions per Session × Time per Session

Or for a content platform:

Engagement = Content Consumed per User × Time per Content Piece

Or the stickiness ratio:

Stickiness = DAU / MAU

Retention:

Day-N Retention = Users active on Day N / Users in that cohort

Always think about retention curves — where is the biggest drop-off? Day 1 (onboarding problem), Day 7 (habit-forming failure), Day 30 (long-term value problem)?

Marketplace Metrics (Uber, Airbnb, DoorDash):

Bookings = Searches × Search-to-Detail Rate × Detail-to-Book Rate × Book-to-Complete Rate
GMV = Bookings × Average Booking Value
Take Rate = Revenue / GMV

Marketplace Liquidity = Supply / Demand (per geography per time window)

SaaS Metrics:

MRR = Number of Customers × Average Revenue per Customer
Net Revenue Retention = (Starting MRR + Expansion − Contraction − Churn) / Starting MRR

Ad Platform Metrics:

Ad Revenue = Impressions × Fill Rate × CPM
CTR = Clicks / Impressions
CPC = Ad Revenue / Clicks
ROAS = Revenue from Ad-Driven Conversions / Ad Spend

The 4-Step Diagnosis Protocol — In Full Detail

Step 1: Clarify the Metric

Before you do anything else, make sure you understand exactly what is being measured and how. Metric definitions are frequently ambiguous, and misunderstanding the metric will send you down the wrong path entirely.

Questions to ask yourself (or the interviewer):

- What is the exact definition? (Example: "Does 'active user' mean logged in, or opened the app, or performed a core action?")
- What is the time window? (Daily, weekly, monthly? Rolling average or calendar-period?)
- Is this a count, a rate, or a ratio? (Each requires a different approach.)
- What is the comparison baseline? (Week-over-week? Month-over-month? Year-over-year? Against a forecast?)
- Is this the raw number or a per-user / per-session metric?

Step 2: Check External vs. Internal

Before diving into product and data analysis, quickly rule out (or rule in) external factors. Experienced analysts always do this first because external factors are common and easy to check.

External factors to consider:

- **Seasonality:** Is this time of year normally high or low? Post-holiday slump? Summer vacation? Back-to-school?
- **Day-of-week effects:** Weekday vs. weekend patterns?
- **Competitor activity:** Did a competitor launch a product, run a promotion, or have a viral moment?
- **Market events:** Economic downturn? Regulatory change? Global event (pandemic, natural disaster)?
- **Platform changes:** Did Apple or Google change their app store policies, notification permissions, or privacy settings (like iOS ATT)?
- **Media coverage:** Positive or negative press about the company?

How to mention this in an interview: "Before I dive into the product-level analysis, I'd want to quickly check whether this is explained by external factors — specifically seasonality, competitor activity, or any platform-level changes. If we can rule those out, then I'd focus on internal causes."

Step 3: Decompose the Metric Mathematically

Take the metric and break it into its component formula (from the decomposition list above). Then for each component, ask: "Did THIS specific component change?"

Worked Example: "TikTok's average session duration dropped 12%."

Session Duration = Number of Content Pieces Viewed per Session × Average View Duration per Piece

Branch A — Number of pieces viewed per session dropped:

- Are users seeing fewer recommended videos? (Recommendation engine issue)
- Are users scrolling less? (UX issue)
- Are users leaving the session earlier? (Where in the session do they leave?)

Branch B — Average view duration per piece dropped:

- Is content shorter on average? (Creator behavior change)
- Are users skipping faster? (Content quality or relevance issue)
- Is there more ad content interrupting? (Ad load change)

Branch C — Both dropped:

- This suggests a systemic issue — possibly a major app update, a content policy change that reduced supply, or a platform bug.

Step 4: Segment and Isolate

Once you've identified which component changed, cut the data by every relevant dimension to find WHERE the change is concentrated. The goal is to go from "session duration dropped" to "session duration dropped specifically for new Android users in the US who joined through paid acquisition."

The Universal Segmentation Checklist:

You should mentally run through this checklist for every diagnosis case. Not all dimensions will be relevant every time, but you should consider all of them.

1. **Platform:** iOS vs. Android vs. Web vs. Desktop
2. **Geography:** Country, region, city, urban vs. rural
3. **User tenure:** New users (0-7 days), mid-tenure (7-90 days), mature (90+ days)
4. **User type:** Free vs. paid, power user vs. casual, creator vs. consumer
5. **Acquisition channel:** Organic search, paid ads, referral, social, direct
6. **Device characteristics:** Phone model, OS version, screen size, connection speed (WiFi vs. cellular)
7. **Time:** Weekday vs. weekend, time of day, before vs. after a specific date
8. **Feature usage:** Users who use Feature X vs. those who don't
9. **Content type:** Video vs. text, short-form vs. long-form, user-generated vs. professional
10. **Behavioral cohorts:** Users who completed onboarding vs. didn't, users who invited friends vs. didn't

The Isolation Process:

Once you find the affected segment, validate that it explains the aggregate change:

"If new Android users represent 25% of our DAU and their session duration dropped 48%, that would account for a 12% decline at the aggregate level (0.25 × 0.48 = 0.12). This fully explains the drop, so I'm now confident the root cause is concentrated in the new Android user segment."

This kind of quantitative reasoning is exactly what separates strong candidates from average ones.

Chapter 5: Simpson's Paradox and Mix Shifts

This concept trips up even experienced analysts. You must understand it deeply because interviewers love to test it, and it appears in real-world data constantly.

What Simpson's Paradox Is

Simpson's Paradox occurs when a trend that appears in several different groups of data disappears or reverses when these groups are combined. The most common version in analytics is the **mix shift**.

The Classic Example

Imagine an e-commerce company reports that its overall conversion rate dropped from 4.0% to 3.7% month-over-month. You investigate and find:

- Desktop conversion: 5.0% → 5.2% (improved!)
- Mobile conversion: 2.5% → 2.7% (improved!)

How can overall conversion drop if both segments improved? Because the **mix shifted.** Last month, 60% of traffic was desktop and 40% was mobile. This month, 45% is desktop and 55% is mobile. Since mobile has a lower base conversion rate, the overall average dropped even though both segments improved individually.

The math:

- Last month: (0.60 × 5.0%) + (0.40 × 2.5%) = 3.0% + 1.0% = 4.0%
- This month: (0.45 × 5.2%) + (0.55 × 2.7%) = 2.34% + 1.49% = 3.83%

Wait — that gives 3.83%, not 3.7%. The actual numbers would be slightly different, but the principle holds. The point is: the overall metric declined not because anything got worse, but because the proportion of lower-performing segments grew.

Why This Matters in Interviews

Interviewers test Simpson's Paradox because:

1. It tests whether you automatically segment before concluding
2. It tests whether you understand that "the average" can be deeply misleading
3. It tests whether you can identify mix shifts as a distinct category of explanation

How to Spot Mix Shifts

Whenever you see an aggregate metric change, always ask: "Could this be explained by a change in the composition of the underlying segments rather than a change in the behavior within segments?"

Specific triggers that suggest a mix shift:

- A rapid growth in one user segment (new market launch, viral acquisition, new marketing channel)

- A change in product mix (more low-price items sold, shift from premium to free tier)
- A change in platform mix (mobile traffic growing faster than desktop)
- Seasonal shifts in user demographics

How to Handle It in an Interview

"Before I conclude that engagement actually declined, I want to check whether this could be a mix shift. If we recently had a large influx of new users from a marketing campaign, these users typically have lower engagement in their first week. The overall average could drop simply because the proportion of new-to-mature users shifted, even if engagement within each cohort is stable or improving. I'd validate this by looking at engagement trends within cohorts rather than at the aggregate."

Chapter 6: Data Quality — The Cause Everyone Forgets

In real-world analytics, a surprising percentage of metric movements are caused by data issues, not actual user behavior changes. In interviews, mentioning data quality as a possible cause (without overemphasizing it) demonstrates real-world experience.

The Data Quality Checklist

Before concluding that a metric movement reflects real user behavior, check:

1. **Logging changes:** Was there a recent update to how events are tracked? Did engineering deploy a new SDK version? Did a third-party tracking tool change its methodology?

2. **Data pipeline delays:** Is the data complete? If the pipeline hasn't finished processing yesterday's data, metrics will appear lower than they actually are.

3. **Bot traffic:** A spike in metrics could be bots. A drop could be bot filtering being applied for the first time.

4. **Metric definition changes:** Did someone change the business logic behind the metric? ("Active user" used to mean "opened the app" and now means "performed a core action.")

5. **Duplicate counting:** Are events being double-counted due to a retry mechanism? Or undercounted because of deduplication changes?

6. **Time zone or calendar issues:** End-of-month metrics can look different depending on whether the month has 28, 30, or 31 days. Time zone changes for daylight saving can

shift events between days.

How to Mention This in an Interview

Don't lead with data quality — it should be a brief check, not your main hypothesis. A good way to bring it up:

"One thing I'd quickly rule out before diving into the analysis is whether this metric movement could be a data artifact — for example, a logging change, a pipeline delay, or a metric definition update. These are surprisingly common in practice and easy to check. Assuming the data is clean, let me walk through the substantive hypotheses..."

PART III: PRODUCT SENSE AND METRIC DESIGN

Chapter 7: Defining Success Metrics

When an interviewer asks "How would you measure the success of X?", they're testing whether you can connect business strategy to data. The answer is never just one metric — it's a coherent system of metrics.

The Metric Hierarchy

Every product or feature needs three layers of metrics:

Layer 1: North Star Metric (one metric)

This is the single metric that best captures the value your product delivers to users. It should be:

- **Aligned with user value:** It goes up when users get more value from the product.
- **Leading, not lagging:** It predicts future business outcomes (revenue, retention) rather than just measuring past ones.
- **Actionable:** The team can influence it through product decisions.
- **Comprehensible:** Everyone on the team can understand and rally around it.

Examples by company type:

- Facebook: Daily Active Users (DAU) — because the product's value is the network, and daily usage indicates the network is healthy
- Spotify: Time spent listening — because it indicates users are finding music/podcasts they love
- Airbnb: Nights booked — because it captures value for both guests (accommodation) and hosts (income)
- Slack: Messages sent by teams — because it indicates the product is integrated into work workflows
- DoorDash: Orders completed — because it represents value delivered to consumers, restaurants, and dashers

Layer 2: Secondary Metrics (2-4 metrics)

These break down the North Star or measure supporting dimensions. They help you diagnose WHY the North Star is moving.

Example for Spotify (North Star: Time spent listening):

- Number of tracks played per user per day
- Session frequency (sessions per user per day)
- Content diversity (unique artists listened to per week)
- Playlist creation and saves

Layer 3: Guardrail Metrics (1-3 metrics)

These are things that should NOT get worse when you optimize for the North Star. They prevent unintended consequences.

Example for Spotify:

- App crash rate (technical quality shouldn't degrade)
- Customer support tickets (user frustration indicator)
- Creator/artist upload rate (platform health for the supply side)
- Subscription cancellation rate (monetization health)

The GAME Framework for Metric Definition

When you encounter a metric definition case, use this structure:

G — Goal: What is the business objective? Growth? Engagement? Monetization? Retention? Be specific about which phase the product is in. A product in growth mode has different success metrics than one optimizing for profitability.

A — Actions: What user behaviors indicate that the goal is being achieved? List the specific actions a user takes when they're getting value from the product.

M — Metrics: Convert those actions into measurable, well-defined metrics. Specify:

- Exact definition (numerator, denominator, filters)
- Granularity (daily, weekly, per-user, per-session)
- Expected baseline and meaningful change threshold

E — Evaluation: How will you track these metrics? What's the monitoring cadence? What would trigger an investigation? What are the known limitations of these metrics?

Worked Example: "How would you measure the success of Instagram Reels?"

Goal: Instagram launched Reels to compete with TikTok for short-form video engagement and to keep users spending time on Instagram rather than switching to a competitor. The goals are (1) increase overall Instagram engagement, and (2) attract/retain younger users who are the primary TikTok demographic.

Actions that indicate success:

- Users watch Reels (consumption)
- Users watch Reels to completion (quality indicator)
- Users create Reels (supply side)
- Users who engage with Reels also continue using other Instagram features — Stories, Feed, DMs (not cannibalizing)
- Users who were declining in Instagram activity start using Instagram more because of Reels (reactivation)

Metrics:

North Star: **Daily Reels Watch Time per User**

- Why: It captures both reach (how many users watch Reels) and depth (how long they engage). Time spent is a better indicator of value than view counts because a "view" can be a 1-second accidental scroll.

Secondary Metrics:

- Reels DAU / Instagram DAU (what % of Instagram users engage with Reels)
- Average Reels watched per session
- Completion rate (% of Reels watched to the end)
- Reels creation rate (daily Reels created per 1000 DAU)
- Share rate (Reels shared via DM, Stories, or external)

Guardrail Metrics:

- Overall Instagram DAU (Reels shouldn't cause users to LEAVE other surfaces)

- Stories daily views (Reels shouldn't cannibalize Stories)
- Feed engagement (likes, comments on non-Reels content)
- Time to first Reel view in a session (are users finding Reels easily?)
- Creator diversity (are Reels dominated by a few creators or broadly distributed?)

Evaluation:

- Monitor daily at the global level, weekly by geography and user cohort
- A 5% sustained increase in Reels watch time would be considered meaningful
- Key limitation: Watch time can be inflated by autoplay — need to distinguish active watching (user chose to stay) from passive viewing (user left phone open)

Chapter 8: Product Feature Evaluation — "Should We Build X?"

These cases test your ability to think like a product-minded analyst. The interviewer wants to see that you consider the user value, the business impact, the risks, and how you'd validate the decision with data.

The PRO-CON-MEASURE Framework

PRO — The Case For Building It

Think about:

- What user problem does this solve? Is it a real, frequent, important problem?
- How many users are affected? (TAM within the user base)
- Does it align with the company's strategic direction?
- Is there evidence of demand? (User research, competitor success, support tickets requesting this)
- What's the revenue or engagement upside?

CON — The Case Against Building It

Think about:

- What is the engineering cost? (Build time, maintenance, opportunity cost of not building other things)
- Could it cannibalize existing features?
- Could it confuse or overwhelm users? (Feature bloat)
- Are there regulatory or privacy concerns?
- Could it be gamed or abused?

- Does it create marketplace imbalances? (For two-sided platforms)

MEASURE — How You'd Validate

This is where you differentiate yourself. Don't just say "build it" or "don't build it." Say:

"I'd recommend running a limited pilot in [specific market/segment] and measuring [specific metrics] over [specific time period]. My success criteria would be [specific threshold]. My guardrails would be [specific metrics that shouldn't worsen]. If the pilot shows [X], I'd recommend full rollout. If it shows [Y], I'd iterate on the design."

Worked Example: "Should YouTube add a 'Watch With Friends' synchronized viewing feature?"

PRO:

- Social viewing is a proven behavior — people already screen-share on Discord and Zoom to watch YouTube together. There's clear organic demand.
- It increases session duration because social accountability keeps users watching longer.
- It differentiates YouTube from competitors (TikTok, Netflix) who don't offer this.
- It could drive engagement with longer-form content, which has higher ad revenue per minute.
- It creates a "social lock-in" effect — your viewing group is on YouTube, making it harder to switch platforms.

CON:

- Engineering complexity is high: real-time synchronization, latency management, audio/video chat integration.
- It may primarily benefit power users while being invisible to casual users — limited reach.
- Content licensing: some content rights holders may have restrictions on synchronized viewing, especially for music and movies.
- It could reduce individual ad impressions if groups of 5 people watch together instead of separately (5 individual sessions → 1 group session).
- Social features can become vectors for harassment, requiring moderation infrastructure.

MEASURE:

- Pilot in one English-speaking market (e.g., Canada) with users who have high social graph overlap.
- Primary metric: Change in total watch time per user among participants.
- Secondary metrics: Session length, content diversity (are they watching different content socially?), D7 retention of the feature.

- Guardrails: Total ad impressions per user (shouldn't drop >5%), content creator view counts (shouldn't drop), abuse reports.
- Duration: 6-week experiment with at least 100K users per arm.
- Decision criteria: If total watch time per user increases by >8% with no guardrail violations, recommend global launch.

My recommendation: Build a limited MVP — synchronized viewing with text chat only (no audio/video). This reduces engineering cost by ~60% while testing the core hypothesis (does social viewing increase engagement?). If the MVP succeeds, layer on audio/video chat in v2.

PART IV: A/B TESTING — FROM DESIGN TO INTERPRETATION

Chapter 9: Designing an A/B Test

A/B test design is one of the most technical case archetypes, but the underlying logic is straightforward once you understand the pieces.

The 7 Components of a Complete A/B Test Design

1. Hypothesis

Always state it formally: "We believe that [specific change] will cause [specific metric] to [increase/decrease] by [estimated magnitude] because [reasoning]."

Bad hypothesis: "The new design will improve the user experience." Good hypothesis: "We believe that adding a progress bar to the checkout flow will increase checkout completion rate by 5-10% because user research shows that 30% of cart abandoners cite uncertainty about how many steps remain as their primary reason for leaving."

2. Randomization Unit

This is WHAT gets randomly assigned to control or treatment. The most common options:

- **User-level:** Each user is randomly assigned. Most common and usually correct. Use this when the feature change affects the individual user experience.
- **Session-level:** Each session is randomly assigned (same user might be in control in one session and treatment in another). Rarely used because it creates inconsistent experiences and makes long-term effects unmeasurable.

- **Device-level:** Useful when users aren't logged in (randomize by cookie or device ID).
- **Cluster-level (geo-randomization):** Randomize by city, region, or market. Use this when there are network effects — e.g., a change to Uber's matching algorithm affects both riders and drivers in the same area, so you can't randomize individually.
- **Page-level or content-level:** Each piece of content is randomly assigned a treatment. Use for testing content ranking or display changes.

The key question: "Could one user's assignment to treatment affect another user's outcome?" If yes, you need a higher-level randomization unit (cluster/geo).

3. Population

Who is included in the experiment? Options:

- All users
- New users only (for testing onboarding changes)
- Users in a specific geography
- Users on a specific platform
- Users who reach a specific point in the funnel

Be specific about inclusion criteria and think about whether your target population affects generalizability.

4. Sample Size and Duration

You don't need to calculate exact numbers in an interview, but you need to show you understand the key factors:

Sample size depends on four things:

- **Baseline conversion rate:** The current value of your metric. Lower baselines need larger samples.
- **Minimum Detectable Effect (MDE):** The smallest effect you'd consider meaningful. Smaller MDEs need larger samples.
- **Significance level (α):** Usually 0.05. The probability of a false positive (detecting an effect that doesn't exist).
- **Power ($1-\beta$):** Usually 0.80. The probability of detecting a real effect when it exists.

Rules of thumb for interviews:

- A typical consumer web experiment needs 1-4 weeks to reach sufficient sample size.
- If the metric has high variance (like revenue), you need more samples.
- If the expected effect is small (<2%), you need a large sample or a long duration.
- Duration should be at least 1-2 full weeks to capture weekday/weekend patterns.

How to discuss this: "I'd run a power analysis to determine sample size. Given our daily traffic of approximately [X], an expected effect size of [Y]%, and standard significance and power levels, I'd estimate we need about [Z] weeks. I'd want at least two full weeks to account for day-of-week effects."

5. Primary Metric and Guardrails

Every experiment needs:

- ONE primary metric that determines the ship/no-ship decision
- 2-4 guardrail metrics that flag unintended harm
- Optionally, secondary metrics for learning

The primary metric should be directly connected to the change you're making. If you're testing a new checkout flow, the primary metric is checkout completion rate, NOT overall revenue (which has too many other influences).

6. Potential Biases and Threats

Always proactively discuss these in an interview. It shows experimental maturity.

- **Novelty effect:** Users engage more with something simply because it's new. The treatment may show a lift in Week 1 that fades by Week 3. Mitigation: Run the test long enough for novelty to wear off (usually 2-4 weeks).

- **Primacy effect:** The opposite — users are initially resistant to change and only engage with the new version after getting used to it. Mitigation: Look at the metric trend over the experiment duration, not just the overall average.

- **Network effects / interference / spillover:** In social products, treated users might influence control users' behavior. Example: if you test a new sharing feature, treated users might share more content that control users then see, improving control metrics. This would underestimate the treatment effect. Mitigation: Use geo-randomization or network-cluster randomization.

- **Survivorship bias:** If the treatment causes some users to drop off, the remaining treated users might look better simply because the weaker users left. Mitigation: Analyze on an intent-to-treat basis (include ALL assigned users, not just those who engaged).

- **Cannibalization:** The new feature might improve one metric by stealing engagement from another surface. Check guardrail metrics on adjacent features.

7. Decision Criteria

Be explicit about what action you'd take for each possible outcome:

- "If the primary metric shows a statistically significant improvement with no guardrail violations, I'd recommend shipping to 100%."
- "If the primary metric improves but a guardrail is violated, I'd investigate the guardrail violation before deciding."
- "If the result is not statistically significant, I'd look at the confidence interval. If the CI includes meaningful positive effects, I might extend the test. If the CI is tight around zero, I'd call it a null result and not ship."
- "If the primary metric worsens, I'd stop the test and investigate."

Worked Example: "Design an A/B test for a new onboarding flow on Duolingo"

Hypothesis: We believe that a redesigned onboarding flow — which lets users choose a specific goal ("Travel conversation in 30 days") rather than just selecting a language — will increase Day-7 retention by 10-15% because goal-setting creates commitment and personalized content from Day 1.

Randomization unit: User-level. New users are randomly assigned upon their first app open. There are minimal network effects in onboarding.

Population: New users who download the app and open it for the first time. Exclude users re-installing (they already have an account). Run globally but analyze by market.

Sample size and duration: Duolingo gets approximately 100K new users per day. With a baseline D7 retention of roughly 25%, an MDE of 3 percentage points (12% relative lift), at $\alpha=0.05$ and power=0.80, we'd need approximately 15K users per arm. With 50K per arm per day, we'd have sufficient sample in 1 day. However, I'd run for 14 days to capture weekday/weekend variation and to measure D7 retention for users who joined on different days.

Primary metric: Day-7 retention (% of users who opened the app on day 7 after install).

Secondary metrics: Lessons completed in first week, time spent in first session, day of first streak achievement.

Guardrails: App uninstall rate in first 24 hours, customer support contacts about confusion, completion rate of the onboarding flow itself (if the new flow is harder to complete, fewer users reach the main product).

Potential biases: Novelty effect is possible — users might engage more with the goal-setting feature initially. I'd compare D7 retention for users who joined in Week 1 vs Week 2 of the experiment to check for temporal stability.

Decision criteria: If D7 retention improves by ≥3 percentage points with statistical significance and no guardrail violations, ship globally. If the lift is 1-3 points, extend the test and add D14 and

D30 retention as secondary endpoints. If the new flow has a lower completion rate (guardrail violation), iterate on the flow design before retesting.

Chapter 10: Interpreting A/B Test Results

Interpreting results is as commonly tested as designing experiments. Here are the situations you need to handle:

Situation 1: "The test is significant. Should we ship?"

Don't just say yes. Check:

1. **Practical significance:** Is the effect size large enough to matter? A statistically significant 0.1% improvement in click-through rate might not justify the engineering maintenance cost of a new feature.

2. **Consistency across segments:** Does the improvement hold across platforms, geographies, and user types? If it only works for iOS users in the US, you need to understand why.

3. **Guardrail check:** Did any guardrail metric worsen? If engagement went up but revenue went down, you need to dig deeper.

4. **Novelty check:** Has the effect been stable over the duration of the test, or was it strong in Week 1 and declining since?

5. **Multiple comparisons:** If you tested 20 metrics, expect 1 to be "significant" by chance. Was the significant metric your pre-registered primary metric?

Situation 2: "Treatment has higher CTR but lower revenue. What do you do?"

This is a very common interview question. The answer depends on understanding the relationship between the metrics:

"This pattern could be explained by several mechanisms. First, the treatment might be attracting clicks on lower-value content or products — users click more but purchase cheaper items or click on non-revenue-generating content. I'd check the distribution of click destinations between control and treatment.

Second, the treatment might be optimizing for engagement at the expense of commercial intent — for example, showing more entertaining content that gets clicks but fewer product recommendations that drive purchases. I'd segment by content type.

Third, there could be a funnel issue — higher CTR brings more users into the purchase funnel, but the downstream conversion might be worse because these are lower-intent users. I'd look at the full funnel: CTR → product page → add to cart → purchase.

My recommendation would depend on the magnitude and the company's strategic priority. If we're in a growth phase prioritizing engagement, the CTR lift might be acceptable even with a small revenue trade-off. If we're optimizing for revenue, I'd reject this treatment or iterate to find a version that lifts both."

Situation 3: "The p-value is 0.08. What do you do?"

"A p-value of 0.08 means we can't reject the null hypothesis at the conventional 5% significance level. However, I wouldn't automatically call this a failure. I'd consider several things:

First, what does the confidence interval look like? If the 95% CI is [-0.5%, +3.5%], the true effect could plausibly be meaningful, and we might be underpowered. I'd consider extending the test.

Second, what was our pre-specified significance level? If we agreed on $\alpha=0.10$ beforehand (which is sometimes appropriate for product experiments), this result would be significant.

Third, how does the practical significance compare? If the point estimate is a 2% lift in a metric that drives millions in revenue, even a suggestive result might warrant further investment.

Fourth, I'd look at the trend over time. If the effect has been growing over the test duration, the test might just need more time. If it's been shrinking, it's likely a novelty effect fading.

My recommendation: don't ship based on this result alone, but also don't kill the idea. Either extend the test to increase power, or run a refined version of the experiment."

Situation 4: "We found a Sample Ratio Mismatch (SRM)"

A Sample Ratio Mismatch means the number of users in control and treatment doesn't match the expected ratio (usually 50/50). For example, if you expected 100K users in each arm but got 95K in control and 105K in treatment, something is wrong with the randomization.

"An SRM is a serious red flag. It means the randomization process is compromised, and the results cannot be trusted regardless of what they show. Even a statistically significant result under SRM is meaningless because the groups aren't comparable.

Common causes of SRM:

- A bug in the assignment logic that correlates with user behavior

- The treatment causing app crashes (users who crash don't get logged)
- Bot filtering applied unevenly
- The treatment affecting sign-up completion (e.g., if randomization happens at sign-up but the treatment makes sign-up harder, fewer users complete treatment assignment)

My recommendation: stop the experiment, investigate the SRM root cause, fix it, and restart. Do not attempt to salvage the results."

PART V: GROWTH, STRATEGY, AND BUSINESS REASONING

Chapter 11: Growth Strategy Framework

Growth cases test whether you can think systematically about what drives a business forward. The key is to be specific and prioritized, not to list every possible idea.

The Growth Equation (AARRR / Pirate Metrics)

Every growth strategy maps to one or more stages of the user journey:

Acquisition → Activation → Retention → Revenue → Referral

For any "How would you grow X?" question, your first move is to identify which stage is the biggest lever. You do this by asking: "Where is the biggest drop-off or the biggest untapped opportunity?"

Acquisition: How do users discover and sign up for the product?

- Channels: organic search/SEO, paid ads (Google, Meta, TikTok), content marketing, partnerships, app store optimization, social/viral, PR
- Key metrics: CAC (cost per acquisition), sign-up volume by channel, channel-specific conversion rates
- Key question: "Which acquisition channel has the best unit economics AND room to scale?"

Activation: What percentage of new users experience the product's core value?

- The "aha moment" — the action that correlates most strongly with long-term retention

- Examples: Facebook — adding 7 friends in 10 days. Slack — a team sending 2000 messages. Dropbox — putting a file in a shared folder.
- Key metrics: % of users who reach the aha moment, time to aha moment, onboarding completion rate
- Key question: "What percentage of users are reaching the aha moment, and what are the barriers?"

Retention: Do users come back after their first experience?

- Retention curves: Day 1 (immediate value?), Day 7 (forming a habit?), Day 30 (long-term value?)
- Key question: "Where is the biggest drop-off in the retention curve, and what do retained users do differently from churned users?"

Revenue: How is the business monetized?

- Subscription: conversion rate from free to paid, ARPU, expansion revenue
- Transactions: order frequency, AOV, take rate
- Ads: impressions per user, fill rate, CPM
- Key question: "Is revenue growth limited by the number of paying users or by how much each user pays?"

Referral: Do users bring in other users?

- Viral coefficient (K-factor): average invites per user × conversion rate per invite
- K > 1 means exponential growth; K < 1 means referral supplements other channels
- Key question: "Is there a natural point in the user journey where sharing is valuable to the user, not just to the company?"

How to Prioritize Growth Levers

In an interview, don't try to cover all five stages. Pick the 1-2 stages that represent the biggest opportunity and go deep on those. Here's how to identify which ones matter most:

- If the product has a strong brand and lots of sign-ups but low retention → **Focus on Activation and Retention**
- If the product has great retention among existing users but slow growth → **Focus on Acquisition and Referral**
- If the product has lots of free users but low revenue → **Focus on Revenue (monetization / conversion)**
- If the product is in a new market → **Focus on Acquisition and Activation**

Worked Example: "How would you grow Duolingo's DAU in Brazil by 30% in 6 months?"

Step 1: Understand the current state. Brazil is a huge market with high smartphone penetration and a population that's interested in learning English for career advancement. Duolingo likely already has significant penetration. A 30% DAU increase is ambitious — it requires both acquiring new users and improving retention.

Step 2: Identify the biggest lever. I'd hypothesize that the biggest lever is **retention**, not acquisition. Duolingo probably already has strong brand awareness in Brazil (language learning is a popular category), but most language learners drop off within the first two weeks. If even a modest improvement in Day-14 retention can be achieved, the compounding effect on DAU over 6 months would be substantial.

Step 3: Specific initiatives (prioritized by expected impact):

Initiative 1 (Retention — highest impact): **Localized social accountability features.** Brazilians have a strong social culture. Launch group learning challenges tied to WhatsApp (the dominant messaging app in Brazil). Users can form study groups with friends and compete on weekly XP leaderboards shared via WhatsApp. Expected impact: +8-12% D14 retention improvement based on benchmarks from social features in other markets.

Initiative 2 (Activation — high impact): **Career-focused onboarding for the Brazilian market.** Instead of generic language selection, new Brazilian users see: "Learn English for [your career goal]." Options: "Job interviews," "Business communication," "Travel." This creates immediate relevance and commitment. Expected impact: +15-20% improvement in users reaching the "aha moment" (completing 3 lessons).

Initiative 3 (Acquisition — medium impact): **Partnerships with Brazilian employers and universities.** Many companies in Brazil offer English training as a benefit. Partner with 10-20 major employers to offer Duolingo Super as a corporate benefit, driving acquisition of highly motivated users who have better retention. Expected impact: +50K-100K new DAU from the corporate channel.

Step 4: Measurement plan. Primary metric: DAU in Brazil. Secondary: D7 and D14 retention by cohort, new user signups by channel, weekly active learner rate. I'd launch Initiative 1 and 2 in parallel (they target different stages of the funnel) and measure impact over the first 8 weeks before committing to Initiative 3.

Chapter 12: Market Sizing and Estimation

Market sizing questions are less common in data science interviews than in consulting or PM interviews, but they do appear, especially at FAANG companies. The skill they test — structured estimation under uncertainty — is broadly useful.

Two Approaches

Top-Down Approach: Start from a large, known number and narrow down through filters.

Example: "How many Uber rides happen in NYC per day?"

Start: NYC population ≈ 8.3 million

- Not everyone uses Uber. Smartphone ownership among adults ≈ 85%. Adults who have used a ride-share app ≈ 40%. That gives us about 2.8 million potential users.
- Of those, daily active ride-share users are maybe 5% on any given day ≈ 140,000 users.
- Average rides per active user per day: approximately 1.3 (some take return trips).
- Total rides per day ≈ 180,000.
- Uber's market share vs. Lyft is roughly 70/30, so Uber rides ≈ 125,000 per day.
- Sanity check: NYC has about 13,000 yellow taxi medallions doing roughly 5 trips each per day (65,000 taxi rides). Ride-share has surpassed taxis, so 125,000 Uber rides seems plausible.

Bottom-Up Approach: Start from a unit and multiply up.

Example: "What's the annual revenue of all coffee shops in San Francisco?"

Start: How many coffee shops in SF? SF has about 870K residents. Rule of thumb: roughly 1 coffee shop per 500-1000 residents in a city like SF that's heavy on coffee culture. Estimate: about 1,200 coffee shops.

- Average revenue per coffee shop per day: maybe 200 transactions × $5.50 average = $1,100/day.
- Annual revenue per shop: $1,100 × 365 ≈ $400,000.
- Total: 1,200 × $400,000 = $480 million per year.
- Sanity check: that's about $550 per resident per year, or roughly $10-11/week on coffee per person. For SF, that's plausible (though probably on the high end — not all residents are coffee shop regulars, but tourists and commuters add volume).

Principles for Estimation

1. **State every assumption explicitly.** The interviewer wants to see your reasoning, not your answer.
2. **Round aggressively.** Don't multiply 8.3M × 0.847 in your head. Use 8M × 85% = 6.8M. Round to 7M if needed.
3. **Sanity-check your final answer.** Does it seem reasonable? Compare to something you know.
4. **Show awareness of uncertainty.** "My estimate is about 125,000 Uber rides per day in NYC, with a reasonable range of 80,000 to 180,000."
5. **Top-down and bottom-up should roughly agree.** If they differ by 10x, you have a bad assumption somewhere.

PART VI: TECHNICAL DEPTH

Chapter 13: Statistics You Must Know Cold

You don't need to derive formulas, but you need to reason about statistical concepts fluently.

Hypothesis Testing — The Full Mental Model

What it is: A framework for deciding whether an observed effect is real or could have occurred by random chance.

The logic:

1. Assume the null hypothesis is true (there is no real effect).
2. Calculate how likely the observed data would be under the null hypothesis.
3. If the data would be very unlikely under the null ($p < \alpha$), reject the null and conclude the effect is probably real.

Critical misconception to avoid: The p-value is NOT "the probability that the null hypothesis is true." It IS "the probability of observing data this extreme (or more extreme) IF the null hypothesis is true." These sound similar but are fundamentally different.

Type I Error (False Positive): You conclude there's an effect when there isn't one. Probability = α (usually 5%). This is like a fire alarm going off when there's no fire.

Type II Error (False Negative): You miss a real effect and conclude there's nothing. Probability = β (usually 20%, since power = $1-\beta = 0.80$). This is like a fire alarm failing to go off when there IS a fire.

The tradeoff: You can reduce Type I errors by requiring stronger evidence (lower α), but this increases Type II errors (makes it harder to detect real effects). In A/B testing, the practical consequence is: a stricter significance threshold means you need larger sample sizes.

Confidence Intervals

A 95% confidence interval means: if we repeated this experiment many times, 95% of the calculated intervals would contain the true effect. It does NOT mean "there's a 95% probability the true effect is in this interval" (a common misconception).

Why CIs matter more than p-values in practice: The CI tells you the range of plausible effect sizes. A p-value of 0.03 tells you the effect is "statistically significant," but the CI [-0.1%, +4.2%]

tells you the effect could be anywhere from slightly negative to meaningfully positive. The CI is more informative.

In interviews, demonstrate CI reasoning: "The point estimate is a 2% lift, but the 95% confidence interval is [0.5%, 3.5%]. Even the lower bound suggests a meaningful improvement, which gives me confidence that this is a real and practically significant effect."

Power Analysis

Power = the probability of detecting a real effect when it exists = 1 - β

The four levers of power (changing any one affects sample size requirements):

1. **Effect size:** Larger effects are easier to detect → need smaller samples
2. **Sample size:** More data → more power → detect smaller effects
3. **Significance level (α):** Relaxing α (e.g., 0.10 instead of 0.05) → more power but more false positives
4. **Variance of the metric:** Lower variance → more power (this is why metrics like "revenue" with high variance require larger samples than "click-through rate" with lower variance)

Variance reduction techniques (advanced, but impressive in interviews):

- **CUPED (Controlled-experiment Using Pre-Experiment Data):** Use pre-experiment metric values as a covariate to reduce variance. If a user's pre-experiment behavior predicts their experiment-period behavior, you can subtract out that expected baseline and analyze only the residual, which has lower variance. This can reduce required sample size by 50% or more.
- **Stratified sampling:** Ensure control and treatment have balanced proportions of key segments (e.g., power users, new users). Reduces variance from between-group differences.
- **Using ratio metrics:** Sometimes per-user-per-session metrics have lower variance than per-user metrics.

Regression — What You Need to Know

Linear regression: Predicts a continuous outcome as a linear combination of features. Use when your target variable is continuous (revenue, time spent, satisfaction score).

Key things to discuss:

- **Coefficients:** "A coefficient of 3.2 on the 'number of friends' feature means that each additional friend is associated with a 3.2-unit increase in the outcome, holding all other variables constant."

- **R^2:** The proportion of variance in the outcome explained by the model. $R^2 = 0.4$ means the model explains 40% of the variance. Important: a low R^2 doesn't mean the model is useless — it might capture the most important relationships even if much of the variance is random noise.
- **Multicollinearity:** When predictor variables are highly correlated with each other, coefficient estimates become unstable and hard to interpret. Diagnose with VIF (Variance Inflation Factor).
- **Omitted variable bias:** If an important variable is left out of the model, the coefficients on included variables may be biased. This is the biggest threat to causal interpretation of regression results.

Logistic regression: Predicts the probability of a binary outcome (churn/not churn, click/not click, convert/not convert).

Key difference: coefficients are in log-odds, not linear units. "A coefficient of 0.5 means a one-unit increase in the feature multiplies the odds of the outcome by $e^{0.5} \approx 1.65$, or a 65% increase in odds."

Chapter 14: Causal Inference — When A/B Testing Isn't Possible

This is one of the highest-value technical topics in interviews because it comes up frequently and most candidates handle it poorly. Interviewers ask about causal inference when they want to know whether you can reason about cause and effect beyond just running experiments.

Why You Can't Always A/B Test

- The treatment already happened (a policy change, a competitor launch, a natural disaster)
- It's unethical to randomize (you can't randomly deny some users a safety feature)
- Network effects make user-level randomization invalid
- The unit of randomization is too large (countries, markets) for sufficient sample size
- Engineering can't build the randomization infrastructure in time

The Causal Inference Toolkit

Method 1: Difference-in-Differences (DiD)

When to use: You have a treatment group (affected by the change) and a control group (not affected), and you have data from before and after the change for both groups.

How it works: Calculate the change in the outcome for the treatment group, subtract the change for the control group. The difference in these differences is your estimated causal effect.

Example: "We launched a new driver bonus program in Dallas but not in Houston. To estimate the effect on ride completion rate, I'd calculate:

- Dallas ride completion change: post-launch minus pre-launch = +5%
- Houston ride completion change: same period = +2%
- DiD estimate: 5% - 2% = 3% attributable to the bonus program."

Key assumption: **Parallel trends** — without the treatment, the two groups would have followed the same trajectory. If Dallas was already on a different trend than Houston, the DiD estimate is biased. Always show the pre-period trends to validate this assumption.

Method 2: Regression Discontinuity Design (RDD)

When to use: Treatment is assigned based on a threshold (score, date, age, etc.).

How it works: Compare outcomes for units just above and just below the threshold. These units are nearly identical except for their treatment status, so any difference in outcomes near the threshold is likely causal.

Example: "We offer free premium access to users with a loyalty score ≥ 80. To estimate the causal effect of premium access on retention, I'd compare retention for users with scores of 78-79 (just below, no premium) to users with scores of 80-81 (just above, got premium). The assumption is that users at 79 and 80 are essentially identical, and any retention difference is caused by the premium access."

Key assumption: Users can't precisely manipulate their score to be just above the threshold.

Method 3: Propensity Score Matching (PSM)

When to use: You're comparing users who voluntarily adopted a feature to those who didn't, and you want to estimate the feature's causal effect.

How it works: Build a model predicting the probability of adoption (the "propensity score") based on observable characteristics. Match each adopter to a non-adopter with a similar propensity score. Compare outcomes between matched pairs.

Example: "To estimate whether using our recommendation engine causes higher purchase rates (vs. the alternative that high-purchase users are just more likely to use recommendations), I'd build a logistic regression predicting recommendation engine usage based on user age, tenure, past purchases, device type, etc. Then I'd match each recommendation user to a non-user with a similar predicted probability, and compare purchase rates between matched pairs."

Key limitation: PSM only controls for OBSERVABLE differences. If there are unobservable factors that drive both adoption and the outcome (motivation, tech-savviness), the estimate is still biased. This is why PSM is weaker than randomized experiments.

Method 4: Instrumental Variables (IV)

When to use: You have an unobserved confounder but can find a variable (the "instrument") that affects treatment but doesn't directly affect the outcome.

This is the most conceptually difficult method. In interviews, it's enough to know when it applies and be able to give an example:

"If I wanted to estimate the causal effect of education on earnings, the problem is that motivation affects both education and earnings (confounding). An instrumental variable approach might use distance to the nearest college as an instrument — it affects how much education you get (people who live closer to colleges tend to get more education) but doesn't directly affect your earnings (living near a college doesn't make you earn more for reasons other than education)."

Chapter 15: Machine Learning in Case Contexts

You'll rarely be asked to build a model in a case interview, but you may be asked when and why you'd use ML approaches, and what tradeoffs you'd consider.

When to Propose ML (and When Not To)

USE ML when:

- The problem is a prediction task with clear labels and training data
- The decision space is too complex for hand-crafted rules
- You need to personalize at scale (recommendations, ranking, targeting)
- Pattern detection in high-dimensional data (fraud, anomaly detection)

DON'T USE ML when:

- A simple rule or heuristic works well enough (e.g., "show most recent first")
- You don't have labeled training data
- The problem is fundamentally a strategy or business decision, not a prediction
- Interpretability is critical and you need to explain every decision (consider simpler models)
- The problem requires causal reasoning (ML finds correlations, not causes)

Key ML Concepts for Case Interviews

Classification — Precision vs. Recall Tradeoff:

This comes up constantly in fraud detection, content moderation, and churn prediction cases.

- **Precision:** Of all the items you flagged as positive, what % actually are positive? High precision = few false alarms.
- **Recall:** Of all the actual positives, what % did you catch? High recall = few missed cases.

The tradeoff is fundamental: raising the threshold to flag fewer items increases precision but decreases recall. Lowering the threshold catches more true positives but also more false positives.

In interviews, demonstrate that you understand how the tradeoff depends on the business context:

- **Fraud detection (financial):** Favor recall — missing a fraud case costs more than investigating a false alarm. But not unlimited recall, because too many false positives create friction for legitimate users.
- **Email spam filtering:** Favor precision — falsely putting an important email in spam is worse than letting a spam email through.
- **Content moderation (child safety):** Favor recall heavily — missing harmful content is catastrophic, even if it means over-flagging some benign content.
- **Medical screening:** Favor recall for initial screening (don't miss cancer), then precision for confirmatory tests (don't give unnecessary treatment).

Recommendation Systems:

Three main approaches:

1. **Collaborative filtering:** "Users who liked X also liked Y." Works well with lots of user interaction data. Cold start problem: can't recommend to new users or recommend new items.
2. **Content-based:** Recommend items similar to what the user has liked before, based on item features. Solves cold start for new items but not for new users. Tends toward homogeneity (filter bubble).
3. **Hybrid:** Combine both. Use content-based for cold start, transition to collaborative filtering as user data accumulates.

In interviews: "I'd start with a hybrid approach. For new users, I'd use content-based features and user profile data (demographics, stated preferences) to provide initial recommendations. As users interact with the product, I'd increasingly weight collaborative filtering signals. I'd also include an exploration component — showing some items outside the predicted preference to avoid filter bubbles and to generate training data for underrepresented items."

Churn Prediction:

This is one of the most common ML use cases discussed in analytics interviews.

Features that typically predict churn:

- Declining usage frequency (strongest signal)
- Decrease in feature breadth (using fewer features over time)
- Support ticket volume
- Failed payment attempts
- Decreased social engagement (fewer messages, shares)
- Time since last purchase
- Contract approaching renewal date

Model selection for churn:

- Start with logistic regression (interpretable, fast to build, good baseline)
- Gradient boosted trees (XGBoost, LightGBM) typically perform best
- Neural networks rarely add value for tabular churn data with <100 features

What to do with churn predictions:

- Don't just predict — take action. A churn model's value is measured by whether interventions reduce churn, not by AUC alone.
- Segment predicted churners by reason: price-sensitive (offer discount), disengaged (send re-engagement campaign), frustrated (proactive support outreach).
- Measure intervention impact with a holdout: randomly withhold intervention from 10% of predicted churners to measure the true causal effect of your intervention.

PART VII: COMMUNICATION AND DELIVERY

Chapter 16: How to Deliver a Case Like a Senior Data Scientist

Technical knowledge is necessary but not sufficient. The best answer, poorly delivered, will lose to a good answer, well delivered. This chapter covers the performance aspect of case interviews.

The First 60 Seconds Set the Tone

When you hear the case prompt, your first instinct will be to start solving immediately. Resist this. The first 60 seconds should be spent demonstrating composure and structure.

Beat 1 (0-15 seconds): Restate and confirm. "So the problem is that our app's daily active users have dropped 15% over the past month, and we need to understand why and recommend a course of action. Is that right?"

Beat 2 (15-30 seconds): Ask 2-3 clarifying questions. "Before I structure my approach, a few quick questions: Is this a gradual decline or a sudden drop? Is it across all platforms or specific to one? And was there any notable product release or external event during this period?"

Beat 3 (30-60 seconds): Verbally present your framework. "Great. Let me structure my approach. I'd break this diagnosis into three main areas: first, I want to determine whether this is a user acquisition problem or a retention problem by decomposing DAU into its components. Second, I want to segment the data to isolate which user groups are most affected. Third, once I've identified the likely root cause, I'd recommend specific actions and a measurement plan."

The interviewer is now oriented. They know your plan, they can follow your logic, and they're confident you won't ramble.

Signposting — The Most Important Communication Technique

Signposting means explicitly telling the interviewer where you are in your framework and what you're about to do. It's the verbal equivalent of headers in a document.

Examples of signposting:

- "I've covered the supply side. Let me now shift to the demand side."
- "There are three possible explanations. Let me walk through each in order of likelihood."
- "Before I give my recommendation, let me quickly address the risks."
- "To summarize where I am: I've identified that the drop is concentrated in new Android users, and my top hypothesis is a recent app update caused performance issues on older devices."

Why this matters: In a 15-20 minute case, the interviewer is processing a lot of information. Signposting helps them follow your logic without effort. If they have to work to keep track of where you are, they'll perceive your answer as less structured even if the underlying logic is sound.

Handling Follow-Up Questions

Follow-ups are not trick questions. They serve three purposes:

1. Testing depth: "Can you go deeper on that?"
2. Redirecting: "That's interesting, but what if the data showed X instead?"
3. Stress-testing: "What if your recommendation fails? What's your backup?"

How to handle them well:

Pause before answering. 3-5 seconds of thinking silence is perfectly fine and shows you're being thoughtful rather than reactive.

If you don't know the answer, be honest and reason through it: "That's something I'd need to look into — I don't have a strong prior on that. But my reasoning would be..."

If the follow-up reveals a flaw in your earlier analysis, acknowledge it gracefully: "That's a good point — I hadn't considered that factor. It would actually change my analysis in the following way..."

Never get defensive. Never argue with the interviewer about what the "right" answer is.

The Close — How to End Strong

The last 60 seconds of your case answer are disproportionately remembered. End with:

1. **A crisp summary:** "To summarize: the most likely cause of the 15% DAU decline is a performance regression in the latest Android app update, which increased load times by 3 seconds for users on devices older than 2 years. This segment represents about 20% of our Android users."

2. **A specific recommendation:** "I'd recommend three actions: first, an immediate hotfix to address the performance regression. Second, a targeted re-engagement push notification to affected users once the fix is deployed. Third, implementing automated performance regression testing in our release pipeline to prevent this from recurring."

3. **A monitoring plan:** "I'd track DAU recovery daily by platform and device type, with an expected full recovery within 2 weeks of the hotfix deployment. I'd also watch Day-7 retention for the affected cohort to determine whether any users were permanently lost."

PART VIII: FULL WORKED CASES — END TO END

Chapter 17: Five Complete Case Walkthroughs

These cases demonstrate the complete thought process from prompt to recommendation. Study them as templates for how to structure your own answers.

Case 1: "Spotify's skip rate increased 20% month-over-month. Diagnose."

Situation (30 seconds): "The skip rate — meaning the percentage of songs that users skip before the song ends — has increased by 20% relative to last month. I'd like to clarify a few things: Is this across all user types (free and premium), and is this global or in specific markets? Also, was there any recent change to the recommendation algorithm or the app's UI?"

[Assume interviewer says: It's across all user types and is global. There was a minor UI update but no algorithm change.]

Problem Decomposition:

"I'd break this into four areas:

First, content quality/relevance: Are the songs being served less relevant to user preferences? This could happen if the content library changed, new music released is less aligned with user tastes, or playlist curation quality declined.

Second, user behavior change: Are the SAME users skipping more, or has the user MIX shifted toward users who naturally skip more? (Simpson's Paradox check.)

Third, UI/UX change: The minor UI update could have inadvertently made skipping easier or changed the skip button's position/size, leading to accidental skips.

Fourth, audio quality or technical issue: Are there playback issues (buffering, audio glitches) that cause users to skip songs that are actually playing poorly?"

Analysis Plan:

"For content relevance: I'd compare skip rates across different content surfaces — Discover Weekly, Release Radar, user-created playlists, album plays. If the skip rate increase is concentrated in algorithmic playlists, it suggests a recommendation quality issue. If it's uniform across all surfaces, it's more likely a user behavior or technical issue.

For mix shift: I'd look at skip rates WITHIN user cohorts (by tenure and usage level) over the two months. If within-cohort skip rates are stable but the proportion of high-skip-rate users grew (say, a big marketing campaign brought in casual listeners), that's a mix shift, not a product problem.

For the UI change: I'd look at the skip rate before and after the UI update specifically, controlling for the time trend. I'd also check whether the skip action method changed (did the proportion of swipe-to-skip vs. button-press-to-skip change?). If swipe-to-skip became easier due to the UI change, we might see more accidental skips.

For technical issues: I'd check skip rate by network type (WiFi vs. cellular) and device type. If the increase is concentrated on slower connections, it could be a buffering issue."

Analysis and Decision:

"Let's say I pulled the data and found that the skip rate increase is concentrated in Discover Weekly and Release Radar (algorithmic surfaces), and is consistent across user cohorts (no mix shift). Within user-curated playlists, the skip rate is flat.

This strongly suggests a recommendation quality issue. I'd dig further into the recommendation pipeline: were the training data updated? Did the feature set change? Was a new model deployed? Even though no 'algorithm change' was announced, underlying data changes (like a new music catalog ingestion) can affect output quality without anyone making an explicit algorithm change.

My recommendation: Investigate the recommendation pipeline for recent changes in training data, feature engineering, or infrastructure. Roll back to the previous model version as a quick test — if skip rates return to baseline within 48 hours, we've confirmed the root cause. Long-term, implement skip rate as a guardrail metric in the model deployment pipeline with automatic alerting."

Edge Cases: "One nuance: a moderate skip rate isn't necessarily bad — it could indicate that users feel empowered to explore and curate. I'd also look at downstream metrics: did the skip rate increase correlate with LESS total listening time (bad — users are frustrated) or MORE total listening time (potentially fine — users are exploring more actively)?"

Case 2: "You ran an A/B test on a new checkout flow. Treatment shows +5% conversion but -3% revenue per transaction. What do you recommend?"

Situation: "So we have conflicting signals — the new checkout flow is converting more users, but each transaction generates less revenue. I'd like to understand: is this revenue-per-transaction metric based on average or median? And what is the checkout flow changing specifically — is it simplifying the flow, or also changing the product selection or pricing display?"

[Assume: It's simplifying the checkout from 5 steps to 3 steps. Revenue per transaction is the mean.]

Problem Decomposition:

"This conflict between higher conversion and lower revenue per transaction can be explained by a few mechanisms:

Mechanism 1 — Incremental converter hypothesis: The new flow is converting users who previously abandoned checkout. These incremental converters might be lower-intent, price-sensitive users who tend to buy cheaper items. The existing converters are buying the same way.

Mechanism 2 — Behavior change hypothesis: The simplified flow is causing ALL users to behave differently — perhaps by removing a step that displayed upsells, cross-sells, or premium options. This reduces revenue per transaction across the board.

Mechanism 3 — Cart composition change: The simpler flow might encourage users to checkout with fewer items (smaller carts) because the speed removes the 'while I'm at it' effect of adding more items during a longer process."

Analysis:

"To distinguish these, I'd segment the analysis:

For Mechanism 1: Look at the distribution of transaction values in treatment vs. control. If the treatment distribution has the same upper end but a new cluster of low-value transactions at the bottom, these are the incremental converters. I'd also compare the conversion rate at different cart value thresholds.

For Mechanism 2: Look at revenue per transaction ONLY among users who would have converted under both conditions (the 'always-converters'). If their revenue per transaction also dropped, Mechanism 2 is at play.

For Mechanism 3: Compare average cart size (number of items) between control and treatment."

Decision:

"The key question is: does the TOTAL revenue (conversion rate × revenue per transaction × traffic) increase or decrease?

If total revenue increases (the +5% conversion more than offsets the -3% revenue per transaction), the new flow is better. The math: 1.05 × 0.97 = 1.019 — a net +1.9% increase in total revenue. This is a clear win, assuming it's statistically significant.

If total revenue decreases, I'd investigate whether we can modify the new flow to recover the revenue loss — perhaps by adding a non-intrusive upsell step back into the simplified flow.

My recommendation: The new checkout flow likely increases total revenue by approximately 2%. I'd recommend shipping it, but adding a lightweight post-purchase upsell or pre-checkout product suggestion to recover some of the per-transaction revenue loss without reverting to the full 5-step flow."

Case 3: "How would you grow Pinterest's male user base by 50% in 12 months?"

Situation: "Pinterest is historically skewed female (~60-70% of users). Growing the male user base by 50% is ambitious. I want to understand: is this 50% growth in male DAU, MAU, or registered users? And does the company have a hypothesis about why male adoption has been lower?"

[Assume: 50% growth in male MAU. The company believes it's a perception problem — men don't think Pinterest is for them.]

Decomposition using AARRR:

"I'd approach this through the growth funnel, but focused specifically on the male user journey:

Acquisition: Are men not discovering Pinterest, or discovering it and choosing not to sign up? Activation: Of men who sign up, are they reaching the 'aha moment' of finding content they love? Retention: Of men who activate, are they coming back?

The '50% growth in male MAU' target means roughly +[X]M male users. I'd need to understand how much of this can come from improving retention of existing male sign-ups versus acquiring entirely new male users."

Strategy (3 initiatives in priority order):

"Initiative 1 (Acquisition — reposition the brand): The biggest barrier is that men don't perceive Pinterest as relevant to their interests. I'd launch targeted campaigns on platforms where men are active (YouTube, Reddit, gaming platforms) showcasing content categories that resonate with male interests: car modification, home gym setups, woodworking, grilling/smoking, tech setups, gaming room design, travel adventure planning. The key is not creating 'content for men' but showing that Pinterest ALREADY has this content — it's a discoverability and positioning problem, not a content gap.

Initiative 2 (Activation — personalized onboarding): When a male user signs up, the onboarding flow should quickly surface content in categories that match male engagement patterns. Right now, the default 'trending' content likely skews toward categories with higher female engagement (fashion, home decor), which would cause a male user to see irrelevant content in their first session and never return. I'd create a male-optimized onboarding flow that surfaces categories like DIY projects, automotive, tech, outdoor adventure, and sports style within the first 3 pins.

Initiative 3 (Retention — social features for male use cases): Research shows men are more likely to engage with utility content (how-to, project planning) than inspirational content. Build features that support the male use case: collaborative boards for group trips, project timelines for DIY builds, shopping lists for builds/projects. Make Pinterest a planning tool, not just an inspiration tool."

Measurement: "Primary metric: Male MAU growth rate (monthly). Secondary: Male signup rate by channel, male D7 and D30 retention, male sessions per week, and male content engagement by category. Guardrails: Female MAU should not decline (ensure we're growing male users, not replacing female users), and content quality metrics should remain stable."

Case 4: "Netflix notices that content completion rates are down, recommendation click-through is up, and subscriber churn is flat. What's happening?"

Situation: "This is interesting because these three metrics seem contradictory. Let me make sure I understand: completion rate means the % of shows/movies that users watch through to the end, recommendation CTR means the % of recommended titles that users click to start watching, and churn is the monthly subscriber cancellation rate. Correct?"

[Assume: Correct.]

Analysis Framework:

"These three data points together tell a story. Let me build a hypothesis that explains all three simultaneously:

Hypothesis: Netflix improved its recommendation thumbnails and titles, which increased click-through (people try more content), but the actual content behind those improved thumbnails isn't always matching user expectations, so they start more shows but finish fewer. Churn is flat because users are still finding ENOUGH content they like — the 'hit rate' of recommendations decreased per-click, but the increased volume of clicks means users still discover good content, just less efficiently.

Let me check this against the data:

If this hypothesis is correct, I'd expect:

- The number of titles started per user per week INCREASED (more clicking)
- The average time before a user abandons a title they don't finish DECREASED (quicker to realize it's not for them)
- The number of titles completed per user per week stayed roughly FLAT (finding the same number of hits, just clicking through more misses)
- User satisfaction surveys (if available) might show slight decline despite flat churn

Alternative hypotheses I'd want to rule out:

Alternative 1: Content library change. Netflix added a large volume of shorter-form content (documentaries, specials, limited series). These have naturally higher CTR (lower commitment to start) but lower traditional 'completion rate' because users watch 2 of 6 episodes instead of all 6. This would be a metric definition issue, not a real quality issue.

Alternative 2: Autoplay changes. If Netflix changed autoplay behavior (auto-starting recommendations more aggressively), CTR would go up mechanically (the user didn't actively choose), but completion would drop (they didn't really want to watch it). Churn might be flat because autoplay still exposes users to content they end up liking some of the time."

Recommendation:

"I'd investigate in this order: First, check whether the recommendation algorithm or thumbnail/title presentation changed in the period when CTR increased. Second, segment completion rate by content type — separate series from movies from specials and see if the drop is concentrated in one category. Third, calculate a 'quality-adjusted CTR' metric: CTR × completion rate. If this composite metric is improving or stable, the system is actually working fine and we're just seeing the natural tradeoff of 'more exploration.'

If the quality-adjusted CTR is declining, I'd recommend tuning the recommendation algorithm to weight completion signals (not just click signals) more heavily, even if it means CTR drops somewhat. A recommendation system optimized purely for CTR will learn to produce clickbait thumbnails."

Case 5: "Design a fraud detection system for a payments platform like Venmo. How would you approach this?"

Situation: "This is a data modeling and ML system design question. Fraud detection is a high-stakes classification problem with significant asymmetry — the cost of missing fraud is much higher than the cost of a false positive, but too many false positives destroy user trust. Let me walk through the design."

System Design:

"I'd approach this in four layers:

Layer 1: Rule-based filters (immediate, real-time) Start with deterministic rules that catch obvious fraud patterns with high precision:

- Transaction amount exceeds user's historical maximum by 10x
- Multiple transactions to the same new recipient within minutes
- Transaction from a device/IP address never associated with this account
- Transaction from a geographically impossible location (user in NYC, transaction from another country 30 minutes later)

These rules are fast, interpretable, and catch the clearest fraud cases. They should be the first line of defense because they run in milliseconds and have very low false positive rates.

Layer 2: ML scoring model (near-real-time, runs on every transaction) A gradient boosted tree model (XGBoost or LightGBM) that scores each transaction with a fraud probability.

Key features:

- Transaction features: amount, time of day, day of week, payment method, recipient characteristics
- User behavioral features: deviation from typical spending pattern, velocity (number of transactions in last hour/day), device fingerprint consistency
- Network features: has the recipient been flagged before? How many connections between sender and recipient? Is this part of a ring?
- Historical features: user's account age, past fraud flags, verification level

Precision-recall tradeoff: I'd set the threshold to achieve approximately 95% recall (catch 95% of fraud) at whatever precision level that implies. For the remaining 5%, we accept some fraud slips through — the cost of blocking legitimate transactions is real (users get frustrated and leave).

Decisions at different score ranges:

- Score > 0.95: Auto-block the transaction. Notify the user.
- Score 0.7-0.95: Add friction — require additional verification (PIN, biometric, SMS code).
- Score 0.3-0.7: Allow the transaction but flag for human review.
- Score < 0.3: Allow without friction.

Layer 3: Network analysis (batch, runs periodically) Some fraud is only detectable at the network level — organized rings where money moves through chains of accounts. Use graph-based analysis to identify suspicious clusters: accounts that are densely connected to each other but sparsely connected to the broader network, accounts created in temporal proximity using similar device fingerprints, and circular payment patterns.

Layer 4: Feedback and model retraining Fraud detection is an adversarial problem — fraudsters adapt to your detection methods. The model must be retrained regularly (weekly or biweekly) with confirmed fraud and confirmed legitimate labels. Monitor model performance metrics (precision, recall, false positive rate) daily with automated alerting if performance degrades.

Evaluation Metrics:

- Primary: Fraud detection rate (recall) — what % of actual fraud do we catch?
- Secondary: False positive rate — what % of legitimate transactions are blocked?
- Business metric: Net loss rate = (undetected fraud losses) / (total transaction volume)

- User experience metric: What % of users experience a false block per month? Target: <0.5%.

Key tradeoff to discuss: Every false positive is a user who tried to send money to a friend and got blocked. That's a terrible experience. Every false negative is money stolen. The optimal threshold depends on the average fraud amount vs. the lifetime value of a user who gets frustrated and leaves. I'd work with the finance team to estimate the dollar cost of each, and set the threshold where marginal cost of false positives equals marginal cost of false negatives."

PART IX: QUICK REFERENCE SHEETS

Metric Decomposition Cheat Sheet

Company Type	Key Metric	Decomposition
Social Media	DAU	New + Retained + Resurrected − Churned
E-commerce	Revenue	Users × Orders/User × AOV
SaaS	MRR	Customers × ARPU
Marketplace	GMV	Supply × Demand Match Rate × Avg Transaction Value
Ad Platform	Ad Revenue	Impressions × Fill Rate × CPM
Streaming	Watch Time	DAU × Sessions/User × Time/Session
Gaming	Revenue	DAU × Payers% × ARPPU

Segmentation Checklist (Run Through for Every Case)

1. Platform (iOS / Android / Web / Desktop)
2. Geography (country, region, city tier)
3. User tenure (new / mid / mature)
4. User type (free / paid, creator / consumer)
5. Acquisition channel (organic / paid / referral / social)
6. Device (model, OS version, connection speed)
7. Time (weekday/weekend, time of day, pre/post event)
8. Feature usage (used feature X vs. didn't)

A/B Test Design Checklist

1. Hypothesis (specific, measurable, with reasoning)
2. Randomization unit (user / session / geo cluster)
3. Target population (all users / specific segment)
4. Sample size and duration (mention MDE, power, baseline)
5. Primary metric (ONE metric for the decision)
6. Guardrail metrics (shouldn't get worse)
7. Biases to watch (novelty, network effects, SRM)
8. Decision criteria (ship / iterate / kill thresholds)

Common Traps to Avoid

Trap	How to Avoid
Jumping to solutions	Always decompose before solving
Forgetting data quality	Ask about logging changes, pipeline delays
Missing mix shifts	Segment before concluding from aggregates
Correlation ≠ causation	Always ask "what else could explain this?"
Only one hypothesis	Present 2-3 ranked hypotheses
No validation plan	Every hypothesis needs a data check
Forgetting guardrails	Always mention what shouldn't get worse
Ignoring second-order effects	Proactively address downstream impacts

---

Study Schedule

Weeks 1-2: Read Chapters 1-3 (Foundations). Do 2 framework drills per day — take a case prompt and write ONLY the SPADE skeleton in under 3 minutes.

Weeks 3-4: Read Chapters 4-6 (Metric Diagnosis). Do 1 full diagnosis case per day. Record yourself and listen back.

Weeks 5-7: Read Chapters 7-10 (Product Sense & A/B Testing). Do 1 metric definition OR experiment design case per day.

Weeks 8-9: Read Chapters 11-12 (Growth & Strategy). Do 1 growth strategy case + 1 market sizing estimation per day.

Weeks 10-11: Read Chapters 13-15 (Technical Depth). Review 1 concept per day (hypothesis testing, CI, power, regression, causal inference, ML basics).

Weeks 12-14: Read Chapters 16-17 (Communication & Worked Cases). Do 3-4 full timed simulations per week. Start with 20 minutes, compress to 15, then 12. Do at least 2 mock interviews with a real person.

Weeks 15-16: Review all quick reference sheets. Do 2 company-specific cases per week targeting companies you're interviewing with. Book 2-3 paid mock interviews on Interviewing.io or Prepfully.

End of The Complete Case Study Mastery Guide

PART X: Product Libraries

FITNESS / WELLNESS (Peloton, Strava, Whoop)

The user motivation curve is the central concept here. Engagement peaks in the first 60-90 days and then naturally decays unless there are strong social hooks or goal structures. External forces that compress workout time include return-to-office reducing morning windows, summer outdoor activity, injury or illness cycles, and holiday travel. Seasonality is extremely pronounced — January is always a spike from New Year resolutions, and July-August is always a trough. Competition for the same time slot is the right framing, not competition for the subscription dollar.

E-COMMERCE / RETAIL (Amazon, Shopify merchants)

The key decomposition is always orders × average order value × items per order. External forces include macroeconomic pressure on discretionary spending, seasonal gifting cycles (Q4 spike, Q1 hangover), and supply chain disruptions affecting inventory availability. User behavior shifts to watch: coupon or discount dependency (users only buy on promotion), category mix shifts (trading down to cheaper items), and channel mix changes (mobile vs. desktop conversion rates differ significantly).

MARKETPLACE (Uber, Airbnb, DoorDash)

These have two-sided dynamics so any metric drop needs to be examined from both supply and demand sides simultaneously. For a rides or delivery metric, demand-side forces include weather, local events, gas prices affecting car ownership decisions, and economic conditions. Supply-side forces include driver incentive changes, competing platforms poaching drivers, and regulatory changes affecting who can operate. The key insight for marketplaces is that a supply shock and a demand shock can look identical in the aggregate metric, but the fix is completely different.

SOCIAL / CONTENT (Instagram, TikTok, Spotify)

Engagement metrics here are heavily driven by content freshness, social graph density, and notification strategy. If a user's social graph goes quiet (friends stop posting), their own engagement drops even if the product is unchanged — this is a network effect working in reverse. External competition is fierce and fast-moving: a new platform pulling attention (TikTok pulling from Instagram) shows up as gradual frequency decline before it shows up as churn. Creator-side health matters enormously — if top creators reduce output, consumption drops within days.

SUBSCRIPTION / SAAS (Netflix, Duolingo, productivity tools)

The central tension is between breadth of content or features and depth of usage. Engagement drops often signal that users have exhausted the content or use cases that motivated their subscription. Key external factors include economic pressure causing subscription audits

Reddit is primarily a social/content platform but with a critical structural difference from Instagram or TikTok: the core unit is the *community* (subreddit), not the individual social graph. This means the standard "social graph went quiet" hypothesis doesn't apply the same way. Instead, Reddit's engagement is driven by community health — if a key subreddit goes inactive, gets banned, or gets flooded with low quality content, users who were there for that specific community disengage entirely even if the rest of the platform is fine. So your hypothesis library for Reddit should always include community-level segmentation first before user-level analysis. External forces that matter specifically for Reddit: news cycles (Reddit engagement spikes around major events), moderation controversies (the 2023 API protest that killed third-party apps is a perfect example of a platform-driven engagement collapse), and the lurker-to-contributor ratio, which is structurally much higher on Reddit than other platforms meaning small changes in contributor behavior have outsized metric impact.

Snap is genuinely a hybrid of social and messaging, which makes it analytically tricky. The core engagement loop is camera-first ephemeral sharing between close friends, not broadcast content consumption. This means Snap's engagement is more sensitive to real-world social activity than almost any other platform — when people are physically together they share less on Snap, which is the opposite of how Instagram works. So seasonality on Snap can actually invert: summer might *increase* engagement if friend groups are apart, or *decrease* it if people are together. The hypothesis you'd generate for a Snap engagement drop should always ask whether the friend graph is still active on both sides of a conversation, because if one person in a streak stops sending, the other stops too. That bidirectional dependency is unique to messaging-first products and creates very fast cascade effects that look like sudden drops rather than gradual declines.

The broader lesson here is that before you apply any library, spend 30 seconds asking: what is the *atomic unit of value* for this product — is it a piece of content, a community, a transaction, a message, or a relationship? That single question will tell you which hypotheses are load-bearing and which are generic filler.

MISSING CATEGORIES FROM THE ORIGINAL LIBRARY

Healthcare / Digital Health (Teladoc, Headspace, Noom)

This category has a unique dynamic that doesn't exist anywhere else: engagement is often *inversely correlated with product success*. If Headspace is working, the user's anxiety decreases and they feel less urgency to meditate — so DAU drops even though the product delivered value. This makes standard engagement metrics deeply misleading. The right framework here is outcomes-based segmentation: are users who engage less doing so because they graduated (healthy churn) or because they gave up (failure churn)? External forces include healthcare seasonality (January mental health resolutions, seasonal affective disorder in winter driving therapy demand), insurance coverage changes affecting willingness to pay, and stigma cycles around mental health that affect acquisition more than retention.

Gaming (Candy Crush, Fortnite, Roblox)

Gaming has the most sophisticated engagement architecture of any category because it's deliberately engineered with variable reward schedules. Engagement drops here almost always trace back to one of three things: content cycle exhaustion (the current season or event ended and the next hasn't launched), difficulty curve miscalibration (users hitting a wall that feels unfair rather than challenging), or social graph collapse (the friend who recruited you stopped playing so you stopped too). The unique external force in gaming is competing game releases — a major title launch from a competitor can pull an entire friend group simultaneously, creating a sudden step-change drop rather than gradual decline. Also important: gaming engagement is extremely age and platform sensitive. Mobile gaming skews heavily toward commute time, so return-to-office patterns affect mobile gaming metrics significantly.

Fintech / Banking (Robinhood, Chime, Coinbase)

The atomic unit here is a financial transaction or portfolio check, and engagement is almost entirely driven by external market conditions rather than product quality. Crypto or equity market volatility directly drives DAU — Coinbase engagement correlates more strongly with Bitcoin price movement than with any product change. This means you should never diagnose a fintech engagement drop without first overlaying market performance data. The unique hypothesis here is *trust erosion* — a single security incident, regulatory action, or high-profile fraud case can suppress engagement for quarters even after the incident is resolved, because financial products have near-zero tolerance for trust violations. Also consider regulatory seasonality: tax season drives massive spikes in financial app engagement that distort quarterly comparisons.

Marketplace for Services (Thumbtack, Fiverr, Upwork)

Different from goods marketplaces because the supply side requires active labor rather than passive inventory. Engagement drops on service marketplaces often originate on the supply side: if skilled freelancers find better-paying opportunities elsewhere (economic boom reducing freelancer availability), buyer request fulfillment rates drop, buyers get frustrated and stop posting, creating a self-reinforcing spiral. The unique edge case here is *trust and review system health* — if review authenticity degrades (fake reviews proliferating), sophisticated buyers disengage first, leaving behind lower-quality demand that further discourages quality supply. This is a marketplace quality death spiral that looks like gradual engagement decline in aggregate metrics.

Enterprise / B2B SaaS (Slack, Salesforce, Notion) —

Engagement drops often have nothing to do with product quality — they reflect internal company changes like layoffs reducing the user base within an account, reorgs changing which teams use which tools, or a new CTO mandating a competitor tool without any grassroots dissatisfaction. This is the critical edge case unique to B2B: *account-level events* drive metric changes that look like product problems. Always segment by account health (logo retention, seats per account, admin activity) before diagnosing product issues. External forces include economic downturns causing seat reductions, merger and acquisition activity consolidating tool stacks, and IT security audits temporarily locking users out. The unique seasonality here is fiscal year budget cycles — Q4 enterprise usage often spikes as teams rush to justify renewals, and Q1 can drop as new budget approvals take time.

Travel / Hospitality (Airbnb, Booking.com, Expedia)

The atomic unit is a trip, which makes this the most seasonally volatile category in the library. Engagement is almost entirely driven by external factors — visa policies, geopolitical stability, fuel prices affecting flight costs, and public health events. The unique hypothesis here is *inspiration vs. transaction split*: many users browse travel products repeatedly without booking, so session-level engagement can stay high while transaction metrics collapse, or vice versa. The edge case to watch is currency volatility — a strong dollar makes international travel cheaper for Americans and suppresses domestic travel simultaneously, creating a mix shift that distorts aggregate booking metrics. Always segment by origin market and destination market independently before drawing conclusions.

Food Delivery (DoorDash, Uber Eats, Instacart)

This sits at the intersection of marketplace and consumer behavior, with extremely strong weather and time-of-day dependencies. Rain increases food delivery orders dramatically — there are internal studies at these companies showing measurable same-day correlation. This means quarterly comparisons without weather normalization are almost meaningless. The unique hypothesis here is *restaurant supply health*: if popular restaurants leave the platform or reduce their hours, demand follows them rather than substituting, because food preference is less fungible than people assume. The edge case is inflation sensitivity — food delivery has extremely visible price markups compared to cooking at home, so during high inflation periods the category sees disproportionate engagement decline as users consciously reduce orders rather than churn entirely.

EdTech (Duolingo, Coursera, Khan Academy)

The central tension is that learning requires sustained effort against natural human resistance to discomfort, which means EdTech engagement is almost entirely dependent on streak mechanics, social accountability, and goal clarity rather than content quality. Engagement drops here almost always trace back to streak loss events — once a user breaks a streak, restarting feels psychologically costly and a significant percentage never return. The unique external force is academic calendar alignment: EdTech that serves students sees dramatic drops during summer and exam periods simultaneously, for opposite reasons (summer is low motivation, exam periods shift users to textbooks and past papers rather than apps). The edge case is *credential value perception* — if the job market softens and users see fewer Coursera certificates leading to employment, completion rates collapse even if learning quality is unchanged, because the extrinsic motivation disappears

Dating Apps (Tinder, Hinge, Bumble)

A user who finds a relationship deletes the app — healthy churn is literally the product working as intended. This means standard engagement and retention metrics are structurally misleading in ways unique to this category. A drop in DAU could mean the product is *more* effective, not less. The right metrics here are time-to-match, message response rate, and date conversion rate rather than raw engagement. The unique external force is demographic pool health — if the gender or age ratio within a market becomes imbalanced (too many men, too few women for example), match rates collapse for the majority group, they disengage, and the imbalance worsens in a self-reinforcing spiral. Seasonality is pronounced: engagement spikes in January (post-holiday loneliness), around Valentine's Day, and in September when people return from summer. The edge case is geographic density dependency — dating apps are fundamentally local network effects products, so a metric drop in a specific city could reflect population migration rather than any product issue.

Streaming / Entertainment (Netflix, Spotify, HBO Max)

The atomic unit is a session or a piece of content consumed, but the engagement driver is almost entirely catalog quality perception rather than catalog size. Netflix has tens of thousands of titles but engagement is driven by whether there are two or three titles the user *personally* wants to watch right now. This is why content release cadence matters more than content volume — a gap between major releases creates measurable engagement troughs even with a massive back catalog available. The unique edge case here is *shared account dynamics*: a single Netflix account might have three to five real users, so account-level metrics mask individual-level behavior entirely. Password sharing crackdowns (as Netflix executed in 2023) create a temporary metric distortion where account count drops but individual user count rises, making engagement per account look artificially higher. Always clarify whether the metric is account-level or device/profile-level before diagnosing anything.

EDGE CASES ACROSS ALL CATEGORIES

Now let me give you the cross-cutting edge cases that apply broadly but are almost always missed:

The Denominator Manipulation Edge Case

Any average metric can improve or worsen purely through denominator changes with no behavioral change whatsoever. If a platform runs a re-engagement campaign and pulls back low-frequency users, the average drops mechanically. If they run a winback campaign specifically targeting high-frequency lapsed users, the average rises. Always ask: did anything change about *who is counted* in the denominator before explaining changes in the numerator.

The Instrumentation Lag Edge Case

This is distinct from simple instrumentation error. Sometimes event logging is correct but there's a processing lag — data from the last few days of a quarter arrives in the next quarter's count. A 25% drop that appeared suddenly at the start of a reporting period should always trigger this hypothesis first. The tell is whether the drop is a step change on a specific date versus a gradual slope over the full period.

The Cannibalization Edge Case

A new feature or product line can suppress engagement on the original metric without any net value destruction. Peloton launching a strength training program might reduce bike workout frequency because users are now splitting time across modalities — total fitness activity is flat but the specific metric you're measuring looks terrible. Always ask whether any new surface or feature launched during the measurement window before concluding there's a problem.

The Notification and Re-engagement Campaign Edge Case

If a platform reduced push notifications or email re-engagement campaigns during the measurement window, passive users who were being artificially dragged back into the app stop appearing in your active counts. This is extremely common because product teams often reduce notification frequency to improve user satisfaction scores, not realizing it will suppress engagement metrics simultaneously. The tell is whether notification volume and open rates changed in the same period as the engagement drop. This edge case is particularly dangerous because it looks like organic disengagement but is entirely reversible through a non-product lever.

The Platform / OS Policy Edge Case

Apple and Google periodically release changes that affect how apps can track, notify, or re-engage users. Apple's App Tracking Transparency (ATT) rollout in 2021 is the canonical example — it suppressed re-acquisition and re-engagement campaigns across almost every consumer app simultaneously, causing industry-wide metric drops that had nothing to do with individual product quality. Any engagement drop that correlates with a major iOS or Android release should immediately trigger this hypothesis. The tell is whether the drop is concentrated among iOS users versus Android users.

The Cohort Maturation Edge Case

This is the subtler version of Simpson's Paradox and is almost never caught without deliberate cohort analysis. As a product matures, early adopters who were highly engaged naturally age into lower engagement as novelty wears off and life circumstances change. Even with perfect new user acquisition replacing churned users, the aggregate engagement average drifts downward because the average tenure of your user base is increasing. This is a structural headwind that affects every maturing consumer product and is almost impossible to reverse without fundamental product reinvention. The tell is whether within-cohort engagement curves are stable but the distribution of cohort ages in your active base is shifting older.

The Currency and Purchasing Power Edge Case

For global products, exchange rate movements can suppress engagement in specific markets in ways that distort aggregate metrics. If a significant portion of your user base is in an emerging market where the local currency depreciated sharply, subscription affordability drops, lower-income users churn, and the remaining base skews toward higher-income users who may have different usage patterns. This shows up as an aggregate engagement change that is entirely a market composition artifact rather than a product problem.

The Competitive Substitution Timing Edge Case

Competitive threats rarely cause immediate churn — they cause *frequency reduction first*. A user who discovers a competitor app typically runs both in parallel for 30 to 90 days before committing to one. During this parallel usage window they appear as an active user in your metrics but their frequency on your platform is falling. By the time they fully churn your metrics have already been absorbing the impact for months. This means frequency drops often *predict* future churn from competitive pressure rather than reflecting current churn, and should be treated as a leading indicator requiring immediate competitive analysis.

HOW TO USE THIS LIBRARY IN AN INTERVIEW

The way to operationalize all of this is to build a three-layer mental checklist that fires automatically for any metric drop question. The first layer is always instrumentation and measurement — spend 60 seconds here and eliminate it cleanly. The second layer is mathematical decomposition — write out the metric as a fraction and ask what could move numerator and denominator independently. The third layer is the domain-specific hypothesis library — spend 90 seconds running through the four or five most common drivers for the specific product type, including the edge cases most candidates miss.

The single biggest differentiator between a 3/5 and a 5/5 candidate is not knowing more frameworks — it's the speed and confidence with which you move through these layers without being prompted. Interviewers are not grading you on whether you eventually get to the right answer. They are grading you on whether your *default mode* is structured, exhaustive, and business-grounded. Every time you wait for a prompt to think of seasonality or external competition, you are signaling that the idea wasn't pre-loaded — it was reactive. Pre-loading is entirely a practice habit, not an intelligence gap.

The practical drill I'd recommend is this: take any consumer product you use daily and spend five minutes asking "if DAU dropped 20% tomorrow, what would my first ten hypotheses be?" Do this for a different product every day for two weeks. After two weeks you will find that your hypothesis generation in interviews becomes almost automatic because you've built pattern recognition across enough product types that the library retrieves itself rather than needing to be constructed from scratch under pressure.

One final meta-point: the edge cases are almost always where interviews are won. Every strong candidate will mention seasonality and mix shift. Very few will spontaneously mention cohort maturation, notification suppression, OS policy changes, or cannibalization from a new internal feature. When you surface one of those unprompted, the interviewer's mental model of you shifts from "solid candidate" to "this person has actually thought deeply about these problems before." That shift is worth more than any individual correct answer in the conversation.